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Applications of AI and ML: Images, Video & Speech — Behind the Scenes



Session Agenda

Exploring How AI Sees, Hears and Impacts Our World

- 1 Async Recap: What You Saw and Tried
- 2 How Does AI "See" and "Hear"?
- 3 Real-World Failures and Limitations
- 4 Critical Issues: Privacy, Ethics and Bias
- 5 Case Studies and Industry Insights
- 6 Future Trends: Multimodal and Generative AI
- 7 Chat With YouTube Video
- 8 Q and A and Takeaways

Async Recap

What You Explored in Visual and Audio AI Applications



Which application surprised or helped you the most during your async exploration?

- **Image:**
 - Category classification
 - Caption generation
 - Object detection and boundaries
- **Video:**
 - Action and event recognition
 - Auto content summaries
 - Object tracking over time
- **Audio:**
 - Speech-to-text
 - Text-to-speech
 - Voice recognition

How AI "Sees" and "Hears"

Understanding Visual and Audio Input for Machines



- **Computer Vision:**

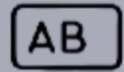
- Images = Pixel grids (RGB values)
- 1080p = 2M+ pixels, 3 colour channels (Red, Green, Blue)
- CNNs detect patterns in images

- **Computer Hearing:**

- Sound = Digital wave samples
- Audio sampled 16k–48k times/sec
- Turned into spectrograms for analysis
- RNNs/Transformers handle sound sequences

Typical AI Vision/Speech Pipeline

From Raw Input to Smart Predictions



Input:

- Image(JPG, PNG)
- Video(MP4, MOV)
- Audio files (WAV, MP3)



Preprocessing:

- Resize and normalise input
- Extract features like edges and textures



Deep Learning Model:

- CNNs for images
- Transformers for audio/video
- Trained on large datasets



Output:

- Labels, captions, or speech transcripts

Real Mistakes in Image, Video & Audio AI

Common Misclassifications



- Confuses objects with similar shapes/colours
- Tags fast-moving games as “fighting”
- Voice assistants triggered by TV or chatter
- Mishears technical terms and rare names

Shows AI's limits in understanding context

Why Do AI Systems Fail?

Understanding The Causes Of Ai Errors



Data Limitations:

Poor or biased training affects results



Environmental Factors:

Lighting, angles, clutter or noise reduce accuracy



Adversarial Vulnerability:

Small input tweaks can fool AI



Edge Case Blindness:

Rare objects or accents often go unrecognised

Discussion: Ever had your phone or assistant get something wrong?

The Business Side of Vision and Speech AI

High Costs, High Value

Hidden Costs:

- Costly manual labelling of images/audio
- Ongoing retraining for accuracy
- GPUs and TPUs for inference
- Scaling for traffic spikes

Commercial Uses:

- Voice and visual search
- Content moderation
- Self-driving tech
- Accessibility tools
- Security systems

High costs make these paid cloud services from major tech firms

Ethics, Privacy and Bias

Key Concerns with Vision and Speech Systems



Identification Risks

Wrong arrests due to facial recognition, mostly affecting minorities



Privacy Concerns

Always-on devices raise concerns about consent and data use



Representational Bias

Models less accurate for non-Western, female or darker-skinned users



Debate Prompt: Should schools, offices, or cities use AI surveillance? What safeguards are needed?

Case Study – Adversarial Example

The "Stop Sign with a Sticker" Attack



Stickers on a stop sign tricked AI into reading it as a speed limit sign

- Small changes invisible to humans
- Caused wrong predictions by the model
- Could lead to self-driving cars missing stop signs
- Shows deep flaws in how neural networks see

These attacks exploit how neural networks process images, not just random errors.

Discussion Prompt:

How can tiny changes fool advanced AI that usually works so well?

Case Study – Bias in Face and Voice AI

When Recognition Fails Certain Groups

Gender Shades Study (MIT)

Landmark research by Buolamwini and Gebru tested commercial facial recognition systems:

- 34.7% error for dark-skinned women
- 0.8% error for light-skinned men
- 43× difference in accuracy

Speech Recognition Bias

Similar patterns emerge in voice recognition:

- Higher errors for African American English
- Struggles with non-native accents
- Misunderstands regional dialects

Impact: Not just tech flaws—these limit access and deepen inequality

Where Is Vision & Speech AI Used?

Real-World Applications Across Sectors



Security and Surveillance

- CCTV, crowd and intruder detection
- Face recognition at airports



Retail Innovation

- Cashierless stores, inventory and shelf monitoring
- Customer behaviour tracking



Healthcare

- Scan analysis (X-ray, MRI, CT)
- Remote monitoring, fall detection



Accessibility Tools

- Captions, screen readers, audio descriptions
- Tools for visual & hearing support



Social Media

- Content moderation, tagging, NSFW and copyright filters
- Engagement-based algorithms



Autonomous Vehicles

- Object detection, traffic signs, lane following
- Obstacle avoidance for self-driving cars

Future Trends in AI

Multimodal Understanding and Generative Power

Multimodal Understanding

New AI models like GPT-4o, Gemini, and LLaVA handle multiple input types at once:

- Understands images and text together
- Processes video with sound
- Reads mixed-media documents
- Combines inputs for better reasoning

Generative Capabilities

AI now creates convincing synthetic media:

- Creates images from text
- Clones voices, generates speech
- Can be generates videos from prompts
- Makes music and sound effects

Next frontier: Real-time, interactive AI that sees, hears and responds

Critical Thinking – Risks and Societal Impact

Challenges Posed by Vision and Speech AI

Deepfakes and Misinformation:

Synthetic media makes fake content easier to create, risking trust and information accuracy

- Fake videos for political use
- Non-consensual synthetic images
- Voice cloning for scams

Mass Surveillance:

Advanced vision AI allows intense monitoring, risking privacy and freedoms

- Facial recognition in public
- Tracking across cameras
- Behaviour prediction

Critical Thinking – Risks and Societal Impact

Challenges Posed by Vision and Speech AI

Labor Displacement:

AI automation of visual and audio tasks puts related jobs at risk.

- AI replacing content moderators
- Automated transcription
- AI for visual checks

Accountability Gaps:

AI errors often raise unclear responsibility between creators, users, and deployers

- Who's liable for AI mistakes?
- False alerts and bias impact
- Unclear responsibility for harm

Group Scenario / Reflection

Balancing Public Safety with Citizen Rights

i **Scenario:** A city plans to use AI cameras for traffic and security.

Possible Benefits:

- Better traffic flow
- Quicker emergency response
- Detects risky driving
- Safer high-crime areas
- Tracks missing persons (AMBER alerts)
- Efficient city service use

Possible Risks:

- Mass surveillance without consent
- Biased impact on some communities
- Wrong IDs and police stops
- Data leaks on people's movements
- Misuse beyond original goals
- Less freedom in public

What rules should protect public rights?

Chat With a YouTube Video

Demo: Chat With a YouTube Video

Use ChatTube to Ask Questions About Any Video

Instructions:

- Go to chattube.io
- Paste a YouTube link (pick an educational, news, or product video)
- Ask: “What is this video about?” “What are the main points?” or “Explain section X in simple words.”
- The app provides AI-generated answers, referencing the video’s content

Chat with any YouTube video

ChatTube allows you to chat with YouTube videos in real-time using AI - ask questions, get summaries, pinpoint key points, translate content, and so much more!

 Chat

Demo: Chat With a YouTube Video



 1

Attention in transformers, step-by-step | Deep ...
3Blue1Brown

 **Brief summary**

The video explores the inner workings of transformers, a critical component of modern AI technologies, particularly in language models. It discusses the attention mechanism, which enables understanding of context in text by associating high-dimensional vectors with tokens. The goal is to refine these vectors for a richer contextual meaning rather than just representing individual words.

Hello! Want to discuss this video's content?

You might want to ask...

How do transformers enhance attention mechanisms in deep learning?

You might want to ask...

How do transformers enhance attention mechanisms in deep learning?

What real-world problems can be solved using attention in transformers?

Why is understanding attention crucial for advancing artificial intelligence?



Messages left: 15

Write a question...

What real-world problems can be solved using attention in transformers?

Attention mechanisms in transformers can address a variety of real-world problems across multiple domains. Here are some examples:

- Natural Language Processing (NLP)**:
 - Machine Translation**: Transformers can translate languages effectively by paying attention to the context and relevance of words in sentences.
 - Sentiment Analysis**: They can analyze text data to determine the sentiment expressed, which can be useful for businesses to gauge customer feedback.
 - Text Summarization**: By focusing on the most important parts of a document, transformers can

Chat with YouTube Video – How Does the AI Work?

What's Actually Happening?

Step	AI/ML Task	What the AI Does	Typical Model/Tech
1	Speech-to-Text	Converts the video's audio into a text transcript	ASR (Automatic Speech Recognition), e.g., Whisper, Google Speech API
2	Text Processing	Cleans, splits, and organises the transcript for retrieval and analysis	NLP, text cleaning, segmentation
3	Retrieval/Augmentation	Finds the most relevant parts of the transcript for a given user query	Information retrieval, embeddings, RAG
4	Question Answering (LLM)	Uses a language model to generate a relevant answer using the retrieved context	LLMs (like GPT-3.5/4, Claude, Gemini)
5	Optional: Summarisation	Summarises sections or entire transcript on request	LLMs, extractive/abstractive summarisers

Q&A Session



Open floor for questions and clarifications.